

Hoang Nam Dang

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SUMMARY

Extensive research experience at the intersection of Human–Computer Interaction (HCI), security, AI alignment, and explainable AI. Proven track record of creating and improving new computational methods to perform dynamic and static analysis of machine learning. Strong ability to collaborate and work in a team environment on multi-disciplinary projects.

EDUCATION

Gannon University, Erie, PA, USA

B.S, Department of Computer and Information Science

- **Thesis:** *Predictive Modeling of Student Transfer Intentions: A Comprehensive Machine Learning Approach with Novel Feature Selection Methodology* (Paper proceeding ETLTC-ICES 2026 conference, University of Aizu, Japan). Co-advised by Dr. Longfei Zhou and Dr. Davide Piovesan. Recipient of J.J Duratz Research Awards 2025 (Rank 1%)
- **Selected publications:** Towards a Serverless Intelligent Firewall: AI-Driven Security, and Zero-Trust Architectures. Proceeding to IEEE CSCloud Conference, New York City 2025
- **Awards:** J.J Duratz Research Awards

SKILLS

- **Programming Language:** Java, Python, C++, Golang, SQL, NoSQL, JavaScript.
- **Framework:** Spring Boot, Fast API, Express, PyTorch, Scikit-Learn
- **Tools:** Docker, Kubernetes.
- **Math:** Linear Algebra, Calculus 1 and 2, Control Theory and Linear System, Probability and Statistics

EXPERIENCE

E3 Company – Erie Technology Incubator

Dec. 2024–May 2025

Software Engineer (on-site)

- Led project team of five that developed a full-stack web application with JavaScript, MySQL, Redis and AWS
- Created a notification and threshold system in the backend that allowed engineers at E3 Company to view updated data in real-time using WebSocket, reducing latency by 20%

PanHealth Inc

Dec. 2024–Jun. 2025

Software Engineer Intern (remote)

- Improved and maintained legacy code in .NET 4/ .NET Web Forms.
- They Developed a MVC pattern that allow user to view and interact with medications.

Department of Computer and Information Science

Jan 2025 - Now

Research Assistant

- Developed an intelligent serverless firewall using LSTM that detected intrusion with the accuracy of 98%
- Optimized LSTM for inference speed, reducing the latency from 3ms to 2ns per invoke, and deployed on a serverless function in AWS ECS, S3 and SageMaker.

Department of Computer and Information Science

Jan 2023 – Aug 2024

Research Assistant

- Researched on explainable AI and supervised machine learning to predict non-matriculating international students at Gannon University using SHAP and Brier score.

Department of Biomedical, Industrial and System Engineer

Jan 2023 – Aug 2024

Smart Manufacturing Research Assistant

- Researched on computer vision techniques to minimize and prevent faults during 3D printing.
- Collected and processed fault-only data to induce noises to the model

Department of Biomedical, Industrial and System Engineer

Jan 2023 – Aug 2024

VR Application Developer

- Researched and developed welding-lathe machine in Unity 3D that helps engineers at ISM practiced in a controlled, simulated environment, reducing operation costs on the machine

PROJECTS

- **Interview Grader:** Built, fine-tuned and deployed an end-to-end, full stack web application in Next.JS and Python that implemented Vision Language Model to ethically assess human expressions. Sponsored by Lockheed Martin.
- **Interns.so:** Participated in OpenAI Hackathon to build a website to match people and build cool stuff. Developed with TypeScript, Python, GPT-OSS 20B and qDrant database.
- **SWETracker:** Built a local-first website that helps candidates store their job applications online without the need to signup

EXTRACURRICULUM ACTIVITIES

- **Volunteer:** Our West Bayfront - Volunteered to help the community around West Erie with civic engagement
- **Cybersecurity Lecturer:** St Paul's Catholic Church - I taught immigrants from Africa, South America and Middle Eastern how to defend themselves against cyber-attacks and phishing emails/ messages.